Deep learning-based Anomaly Detection on Surveillance Videos: Recent Advances

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***Abstract*—Surveillance cameras number keep increasing from year to year, in 2017 alone estimated 106 million new CCTV installed. Surveillance cameras are installed to widen surveillance coverage in order to detect anomalies. Conventionally, detecting anomalies through surveillance cameras are using human observers. The increasing number of surveillance cameras installed is increasing the need for an automatic system to replace unreliable human observer. Thus an intelligent computer vision-based system to detect anomalies need to be developed. The goal of this system is to provide a warning to the first responder accurately and as fast as possible while the system is running all the time. The faster the response of the first responder, the better chance of an anomaly to be resolved.**

Deep learning based method has been proposed to solve anomaly detection on videos. Anomaly detected is composed of a set of violent and non-violent crimes with high impact in society. To increase the accuracy of previous system to detect anomalies in condition that resembles real life situation, multiple processes need to be combined. This paper will review various method used to increase the performance of a deep learning based action recognition on videos. To capture temporal information effectively the model need to be able to use multimodalities to detect motion, incorporating long-range temporal structure, and numerous deep learning architectures with various characteristics.

Keywords—CCTV, anomaly, deep learning, untrimmed, weakly supervised, action recognition.

I. INTRODUCTION

The number of CCTVs keeps increasing from year to year, in 2017 alone estimated 106 million new CCTV installed [1]. The installation of CCTV in public area is used by law enforcement to widen surveillance coverage in order to prevent crimes perpetrated or to detect an accident. The growth of CCTV installation is because of the high correlation between the decreasing number of crimes perpetrated and the installed cameras [2]. One problem that emerges from the increasing number of installed CCTV is the high amount of data needed to parse through by the operator manually. Human capabilities to detect anomalies decreased around 18% in observing four different monitors simultaneously compared to a single monitor, even when extrapolated linearly the capability to detect anomalies will plummet significantly with the increasing number of monitors to observe [3]. Lower capability to detect anomalies will translate to human observer detecting the anomalies far too late or even missed the anomalies altogether. The faster the response of the first responder is crucial to resolve the anomalies that happened [4]. In order to achieve faster response time, an automated system that can observe multiple monitors simultaneously and able to detect the anomalies need to be developed.

One idea is to develop a system based on a computer vision system. The rapid growth of computing system promotes the rapid growth in computer vision research [5]. Multiple methods have been developed in computer vision research on the field of object detection, recognition, and tracking. General step of computer vision processing consisted of two main steps: feature extraction and classification. Anomalous event is defined as a pattern or behavior that is different from the majority of the event. Using this definition as the basis of anomaly detection, one of the most popular method is detecting an event as a two-class classification, an event as a normal event or an anomalous event. Even though this approach seems optimal, all events differ from the learned normal events are going to be detected as an anomaly [6]. This approach will cause a high amount of false alarms because there is a large amount of variation of normal events. One strategy developed to circumvent this problem is by defining a specific anomaly to detect and to train a classifier trained with both anomalous and normal videos. Some of the research tries to detect one specific anomaly such as stealing [7], traffic accident [8], assault [9], or even multiple anomalies at once [5].

Previous state of the art researches [5] are using a deep learn based method to detect multiple anomaly classes on untrimmed video. This research also introduces a dataset that consisted of untrimmed anomaly in real-life surveillance cameras. Current research focus is increasing the accuracy of previous research using segment-based sampling to detect long range temporal structure efficiently [11], techniques to automate dataset annotation, various modalities as inputs to incorporate both temporal and spatial features of the video, and finding the best architecture with lowest computation needed.

II. RELATED WORK

In the field of computer vision one of the most challenging problem is anomaly detection, mostly on the difficulty in extracting a specific feature which correlates to a specific event. Video encodes both spatial and motion information instead of only spatial information on an image. Video analysis using both modalities provides much more information to detect a particular action, but the process to extract additional information requires significant resources. It is important to design a method that can extract information to represent information from videos effectively with a reasonable time frame.

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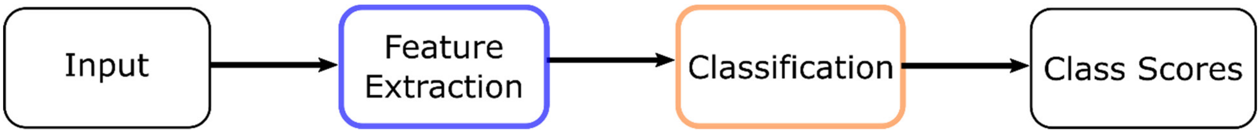

Fig. 1. General processes to recognize an action from video

Several researches have attempted to classify anomaly on videos using handcrafted features. One of the proposed methods to detect anomalies is by using Edge Oriented Histogram (EOH) and Multi-layer Histogram of Optical Flow (MHOF) to describes appearances and movement respectively [12]. Another method utilizes the change in temporal pattern by calculating Markovian differences from the local pattern, while the time scale used is modeled globally [6]. Gaussian Mixture Model (GMM) is used to detect vehicles from video and then used mean shift to calculate position, speed, and direction to determine an accident will occur or not [13].

Besides detecting anomalies in a specific criminal activity, a general approach to detect anomaly is by using spatio-temporal features detector to extract local descriptor of a normal event and training a classifier to acquire the global representation of normal event. Between all of the hand-engineered method, iDT generally regarded as the best method despite the large cost of computation needed [14]. All these methods, when trained using dataset of normal videos, will detect low probabilities pattern as anomalies.

Recently with the rapid growth of computing system, deep learned based method had been developed to learn features and classify an action in videos. Convolutional Neural Network is the first deep learned based method in general image processing. Variants of CNN have been developed to solve image processing problem in video. A 3D CNN uses a 3D kernel (2D filter deployed along the time axis) to be able to extract information from both spatial and temporal dimensions [15]. This method is used as the basis in classifying anomalies on UCF-Crime dataset [5]. Another strategy is utilizing recurrent structure to extract temporal information, which known as Long-Short Term Memory. Another research used CNN to extract spatial feature and then aggregates the results using a variant of LSTM to detect violent events in videos [9]. This research detects all aggressive behaviors as a violent event.

State of art results in video action recognition is achieved using a two-stream based network [16]. Two data streams consisted of individual frames (spatial features) and flow (temporal features) is trained independently with separate deep learn architecture and then the results are aggregated to get the prediction score.

III. Methodology

In this section, we describe multiple methodologies to train deep learning network for detecting various classes of action from videos.

A. Dataset and Evaluation

Performance of the action recognition methods is usually compared using some particular datasets depending on the purpose of the network. Typically, a network is built according to a specific goal, such system won't be able to generate a solution that can used to solve problems in multiple scenarios. Progress of action recognition method are followed by the increasing complexities of the problem to solve. The complexity of a dataset is usually indicated by the resemblance of its contents to reality.

TABLE I. Various Trimmed and Untrimmed Dataset

Dataset Names	***Videos Number***	***Videos Duration***
UCF-101	13320	2-5s
HMDB-51	6766	2-3s
UCF-Crime	1900	3-5 min
ActivityNet	19994	5-10 min
THUMOS' 14	4274	4-5 min

One kind of dataset is a trimmed dataset, this dataset composed of short clips that already cropped of specific actions. From table 1, HMDB-51 [17] and UCF-101 [18] datasets are a trimmed dataset because the content of both dataset contain a well cropped clips of actions. The other kind of dataset is an untrimmed dataset, built based on real-life situation, the actions usually occur in a small portion of the clip. These untrimmed dataset that resembles real-world situation (ActivityNet [19], annual THUMOS [20], and UCF-Crime [5]) has appeared to train action recognition method to real world scenarios.UCF-Crime [5] is a large-scale dataset consisted of 128 hours total of untrimmed real-world surveillance videos. This dataset is composed of 1900 videos with 13 anomalies including Vandalism, Shoplifting, Stealing, Shooting, Robbery, Fighting, Explosion, Burglary, Road Accident, Assault, Arson, Arrest, and Abuse.

All of the methods is evaluated using a frame-level receiver operating characteristic (ROC) curve and then the area under the curve (AUC) will be calculated to get the evaluation metric. The higher the value of AUC implies better accuracy as well as higher robustness of the method.

B. Preprocessing

Multiple techniques are generally used to augment training data. Horizontal flipping is flipping the frames used in training horizontally, effectively doubling the amount of training data. Another technique is using cropping, first is corner cropping by extracting regions selected from only corner or the center of an image. Another technique is multi-scale cropping technique. One example of this techniques is with an input size of 256x340, cropped regions are randomly selected with various size combination, the cropped regions will be resized to 224x224 to train the network. With these various techniques, the amount of data will be multiplied to get a better deep learning network training methodology.

C. Sampling Method

In real-life scenarios, an anomalous instance may only take up a small part of a whole video, with a random duration and starting time while the majority of the video is a normal instance. Another problem in a long untrimmed video is how

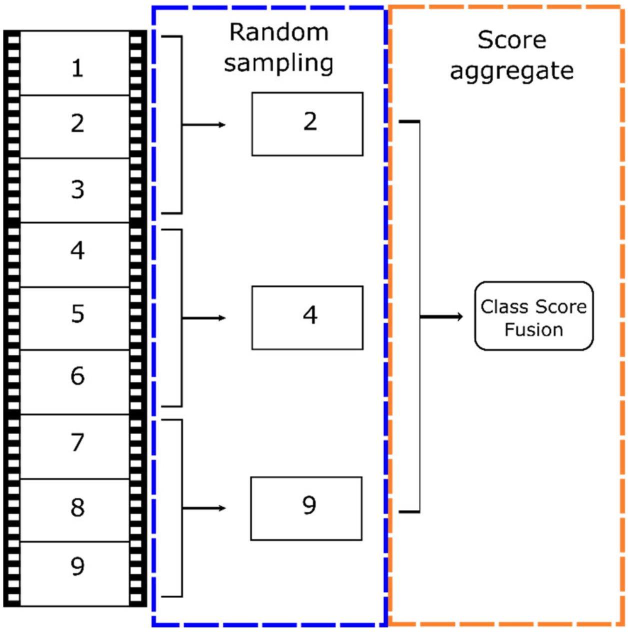

Fig. 2. Random short snippet selected from each segment in TSN framework

to integrate not only the appearances and short-term motionbut also the long-range temporal structure that indicates an anomaly from a normal event. A model trained using data with short-term information (single frame) will try to use all image data to recognize action. One example is a model trying to recognize an act of "Diving", a single frame input will train the network to look for water under the diving platform. A model trained using temporal information focused on the wave motion caused by the act of diving, while a temporal model with long-term information will focus humans on the video. This result suggests a model encoded with long-range temporal structure will perform better in action recognition.

Temporal segment network (TSN) framework can be used to solve the previous problem. This framework basically uses a segment-based sampling instead of working on a single frame or a short frame stack. This method chooses a random snippet from the segmented videos to generates a snippet-level prediction. All the snippet-level predictions will be aggregated into video-level prediction. Video-level prediction using snippet-level aggregates is able to capture long-range information over the entire video, although the method does not process the entire video. Optimal model parameters are updated while training the deep learning network.

D. Dataset Labelling

Training method for a deep learned based method involves using annotated videos. Ideal annotation in anomaly detection consisted of labels of the anomaly and time when the anomaly occurred. Given enough computation power and time in training a deep learning network, a larger dataset will generate better results. While a large dataset is preferable, creating a large dataset with annotation requires a high cost of human labor, especially in anomaly detection in untrimmed videos. Generating labels and times of each anomaly manually is both tedious and difficult. One method to circumvent the need to annotate time is to provide anomaly

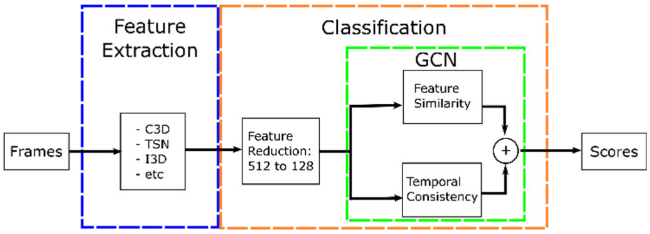

Fig. 3. General diagram of GCN optimization process

label on video-level only. Instead of relying on manuallyannotated time and duration of an anomaly, this weakly supervised method requires an algorithm to predict the anomaly automatically.

Several methods have been developed to solve the prediction problem. One method is by considering the anomaly detection as multiple-instance learning (MIL) task, and the other method is by considering anomaly as a noisy label to be cleaned (GCN). In Graph Convolutional Label Noise Cleaner (GCN) a wrongly annotated video is considered as a noisy label, to obtain a correct anomaly scores automatically. This method is correcting low-confidence (high variance) anomaly scores using a high-confidence (low variance) scores. This technique is alternately training between the action recognition classifier and noise cleaner network. Rough snippet-level scores from the classifier are used in noise cleaner to be corrected. GCN also utilizing video characteristics to help correct the label, which is called as feature similarity and temporal consistency. Temporal consistency means anomaly snippets generally appear close in time, while feature similarity means anomaly snippets generally share similar characteristics.

E. Architectures

The architecture of a deep learning network describes the specific arrangement, size, width, depth, and type of layers and nodes. Variations of these component various network structure have been developed to be able to extract features and classify depending on the purpose of the network. Compared to deep learning network to classify based on images, video-based network need to incorporate temporal information. Without temporal information, the motion of an action cannot be differentiated, i.e. the act of opening a door is similar to closing a door without temporal information. Various characteristic of the specific architecture is specifically tailored to face the challenge in videos action recognition. The usage of temporal motion information increases the accuracy of action recognition in videos. These accuracy improvements suggest a positive correlation between temporal motion information and action recognition for videos.

Main differences in recent development of video-based deep learning network are the myriad of techniques to include temporal motion to help recognize an action; whether using recurrent layers to propagate information across time [21], using convolution layers with additional temporal dimension, or using motion-based flow as input to the network [22]. Multiple types and combination of these techniques are crucial in finding the network with better performances in action recognition in videos.

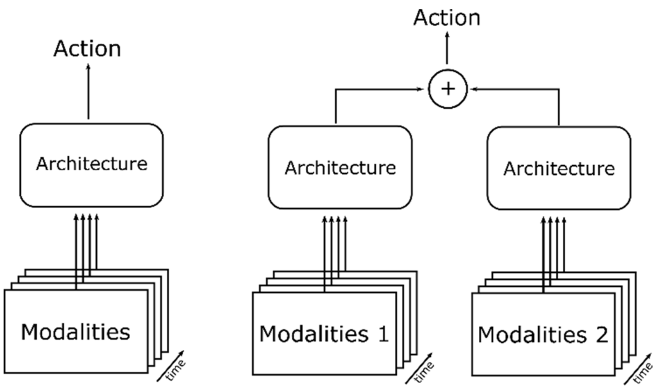

Fig. 4. General single stream (left) and two-stream (right) network

1) ConvNet 3D (C3D) [15]

C3D is one of the first research in trying to incorporate temporal motion information in increasing accuracy in videos action recognition. This research directly trying to encode temporal structure by using 3D convolutional network (ConvNet) instead of 2D ConvNet. 3D ConvNet is achieved by adding additional dimension into $d \times d$ (d correspond to spatial size) 2D ConvNet into $d \times d \times l$ kernel. An additional dimension is added to include temporal dimension where l corresponds to the number of frames used as input.

2) Two-Stream Inflated 3D ConvNets (I3D) [23]

The current state of the art deep learned architecture is using optical flow to capture motion information from video frames and then processing both position and motion information. Optical flow is used to describe the motion of an object to recognize actions. The algorithm to extract optical flow produces separated horizontal and vertical components to capture motion created by an object and then blur the results using a Gaussian filter. Two data streams consisted of individual frames (spatial features) and flow (temporal features) is trained independently with separate deep learn architecture and then the results are aggregated to get the prediction score

While a single-stream C3D can learn temporal information directly from RGB frames sequences, the performance can be improved by including another flow stream. This algorithm models temporal structure by using 2 types of inputs, single RGB frame and a stack of 10 previously computed optical flows. Both modalities are used as input of 2 duplicate ConvNet pre-trained on ImageNet. Besides using 3D Convnet, this architecture also used inflated filters and pooling kernel to include temporal structure. This is achieved by repeating weights of 2D filters from pre-trained ImageNet models along the time dimension.

3) R(2+1)D [14]

One of the main problem with 3D ConvNet models is that the models have larger amount of parameters compared to 2D ConvNets because of the additional kernel dimension, and this makes them harder to train. Another approach in approximating 3D convolutions is decomposing the spatial and temporal processing by using 2D convolutions followed by 1D convolution. The decomposition process provides two advantages. First, additional ReLU layer is added between 2D and 1D convolutions increasing nonlinearities of the network despite the number of parameters is the same. Second, the decomposition of 3D convolutions provides a better optimization process, which in turn produces a lower training error over equivalent network. Utilizing lower training error, depth of the architecture can be increased (8 convolution layers in C3D into 34 ResNet layers in R(2+1)D).

4) Hidden 2 Stream [24]

Optical flow extraction using traditional method (TV-L1, Farneback's) needs high computational cost and requires significant storages. The accuracy and computational load of optical flow highly depend on the algorithm used, generally TV-L1 generate the most accurate optical flow with the heaviest computational load, followed by Farneback's respectively. Traditional optical flow estimation requires a lot of computation time and storage space compared to the whole architecture. This traditional flow extraction method is one of the main bottlenecks in current two-stream approach. Various methods have been proposed to solve the flow extraction.

In this research, an end-to-end deep learning approach, called MotionNet, is used to learn optical flow to avoid costly computation and storage. MotionNet is a special network trained using an unsupervised loss functions with the goal of generating flow from a set of consecutive frames. MotionNet then placed before a typical temporal stream network, treating MotionNet as a traditional flow estimator. This research fused the two streams' scores with a ratio of spatial to temporal stream as 1:1.5. This CNN based optical flow estimator is able to compute much faster results compared to traditional TV-L1 method (120 fps compared to 14 fps on TitanX)

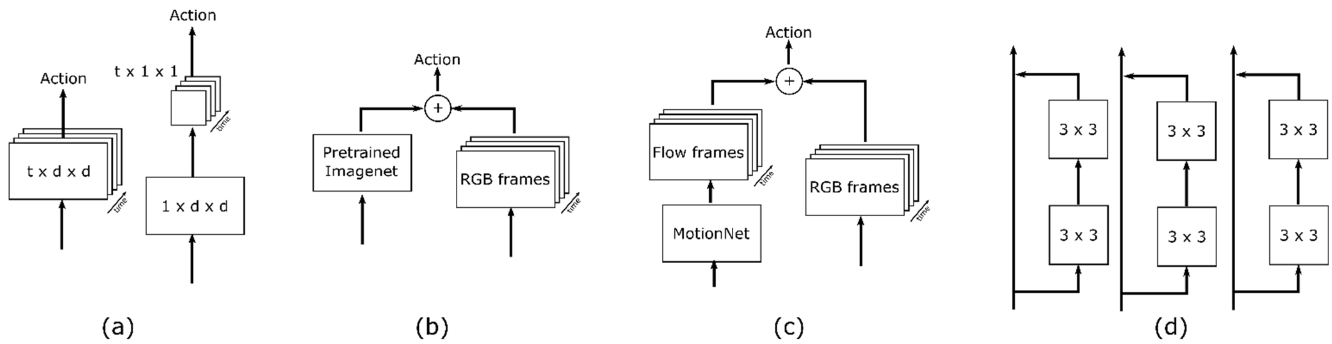

Fig. 5. (a) 3D conv. (left) Decomposed R(2+1)D conv. (right) (b) T3D (c) Hidden 2 Stream (d) Multi-fiber architecture

TABLE II. ACCURACY AND SPEED OF TSN ARCHITECTURE USING DIFFERENT COMBINATION OF MODALITIES ON UCF-101 [11]

Modalities	*Accuracy (%)*	*Speed (FPS)*
RGB+Flow	94.9	14
RGB+Flow+Warp	95.0	5
RGB+RGB Diff.	91.0	340

5) Temporal 3D ConvNets (T3D) [25]

Another approach to model temporal motion is by omitting image flow entirely. This research utilizes supervision transfer across architectures (from 2D network into 3D network) to increase training efficiency. This method can be used to transfer information across modalities too, not limited to transferring information between RGB models only. Transfer learning between cross architectures, pre-trained 2D ConvNets to 3D ConvNets, is used to avoid the need to train 3D ConvNets from scratch. This method leverages pre-trained 2D ConvNets on popular ImageNet dataset to learn feature representation from 3D ConvNets architecture. In training process, parameters of pre-trained 2D ConvNets is changed, so the architecture only needs to learn the parameter of 3D ConvNet while trying to classify an action from videos.

3D ConvNets architecture used in this research consisted of 3D filters (instead of 2D filters) and pooling kernels able to integrates variable temporal depth information over both shorter and longer temporal ranges. The network dubbed as TTL is consisted of several 3D Convolution kernels with variable parameters and a single 3D pooling layer. The depth parameters of 3D Convolution kernels ranges between depth of the input data. This variability of TTL able to captures the short, mid, and long-term data, that contain important information not captured when videos is processed with a network with fixed temporal depth parameter.

6) Temporal Segment Network (TSN) [11]

Most of the above deep learning-based methods on video action recognition focused on the problem of including temporal information in the network. The heavy computation needed to extract optical flow creates a major bottleneck in current two-stream approaches. To solve this problem various methods have been proposed to capture motion information from videos other than using traditional optical flow extraction, one method highlighted in this research is using RGB differences as an approximation of motion. Besides sampling method, this research also compares different modalities to capture temporal information.

From table II can be observed the result of network utilizing optical flow yields the better results, but this method severely slows down the testing speed; 14 fps for flow modality and 5 fps for warped flow modality. These results, processed using a single TitanX, are not desirable in a real-time video processing, with generally acceptable frame rate for a real-time system is at least 30 fps. Using RGB differences to model motions from an object from videos yields a much faster testing speed (340 fps) while still having better result compared to a single stream network.

7) Multi-fiber (MF-Net) [26]

Another research in improving the efficiency of 3D ConvNets, maintaining close to state-of-the-art accuracy while lower computational load, is by using the large input tensor of 3D ConvNets. A sparsely connected architecture, known as Multi-Fiber network, is proposed with each architecture unit consisted of lightweight 3D convolutional

TABLE III. PERFORMANCES OF VARIOUS ARCHITECTURES ON ACTION RECOGNITION

Architectures	*Pretraining dataset*	*Trimmed*		*Untrimmed*	*Total parameter* [23]	*FLOPs* [26]
		UCF-101 (%)	HMDB-51 (%)	UCF-Crime (%)		
C3D [15]		82.3	51.6		79M	38.5G
MIL + C3D [10]	Sports-1M			75.41		
GCN + C3D [10]	Kinetics			81.08		
GCN + TSN - RGB [10]	Kinetics			82.12		
GCN + TSN - Flow [10]	Kinetics			78.08		
I3D – RGB+Flow [23]	Imagenet + Kinetics	98.0	80.7		25M	107.9G
R(2+1)D - RGB [14]	Kinetics	96.8	74.5		63.6M	152.4G
R(2+1)D - Flow [14]	Kinetics	95.5	76.4			
R(2+1)D - RGB+Flow [14]	Kinetics	97.3	78.7			
TSN - RGB+Flow [11]	Kinetics	94.9	71.0			3.8G
TSN - RGB+RGB Diff. [11]	Kinetics	91.0				
MotionNet + TSN [24]	Imagenet + Kinetics	93.2	66.8			
MotionNet + I3D [24]	Imagenet + Kinetics	97.1	78.7			
T3D [25]	Kinetics + YouTube8m	91.7	61.1			
T3D+TSN [25]	Kinetics + YouTube8m	93.2	63.5			
MF-Net-RGB [26]	Kinetics	96.0	74.6		8.0M	11.1G

networks sparsely connected architecture that is independent from each other similar to network of multiple fibers. The sparse connection in this architecture reduces the computational cost as much as the number of fibers. This research also introduces a multiplexer module to improve information flow across fibers. The multiplexer module is attached at the beginning of each residual block. This module able to redirect information between parallel fibers to share representation from input data, and the overall capacity of the model is increased.

IV. Conclusion and Future Works

Recent advances in action recognition on videos show a promising direction in application in real-world scenarios. We show the existing algorithm to recognize an action on a scenario that resembles real-world condition and discuss the advantages and disadvantages. The results suggest that performance increase is affected by pre-training. Pretraining for an action recognition network on videos requires both large datasets and expensive computation to extract temporal information from videos. Various preprocessing has been developed to increase training data from a known dataset. Another method to increase training data is by using unsupervised or weakly supervised from additional data. Sampling method is also used to acquire temporal information in a long untrimmed video.

These factors will be used as a major consideration in choosing a specific algorithm for real-time applications. The future scope of research of anomaly detection on real-world problems (surveillance videos) is focused on finding an accurate algorithm and able to generate prediction using real-time data. This will be achieved by focusing research on generating more training data (pretraining, unsupervised training), exploring modalities to capture more data (temporal flow, RGB differences), different method sampling to capture anomaly events, and exploring various deep learning architecture architectures.

Acknowledgment

This work is funded by the Rekognisi Tugas Akhir (RTA) Program of the Research Directorate of Universitas Gadjah Mada. Any opinions, findings, and conclusions or recommendations expressed herein are those of the authors and do not necessarily reflect those of the Research Directorate of Universitas Gadjah Mada.

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